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Spectral Signatures of SCS from Radio Observations

Abstract

We incorporate constraints on superconducting cosmic strings (SCS) from radio observations (arXiv:2305.09816, 2023) into the UQFF framework. The $[\text{SCm}]$ emission mechanism from SCS produces power spectral density:

$$P(f) = \frac{G\mu \cdot c^2 \cdot I^2 \cdot f}{4\pi d^2} \cdot [\text{SCm}]_{\text{emission}}$$

where I is the supercurrent, d the luminosity distance, and $[\text{SCm}]_{\text{emission}} = \exp(-[\text{SSq}] \cdot 13/26), = 0.7483$ at level 13. The model predicts FRB-like bursts from SCS cusps and kinks, with detectable flux densities at ∼GHz frequencies for nearby ($d \lesssim 1$ Gpc) strings with $G\mu \gtrsim 10^{-8}$. Alignment: 95.00%.

1. Introduction

Fast Radio Bursts (FRBs) remain among the most enigmatic transient phenomena in radio astronomy. While magnetar models explain some repeating FRBs, the origin of one-off FRBs is debated. Superconducting cosmic strings provide a natural mechanism: electromagnetic radiation from cusps and kinks on strings carrying persistent supercurrents.

The UQFF framework connects SCS emission to the $[\text{SCm}]$ vacuum condensate, providing a unified model for both string stability (PAPER_1116) and electromagnetic emission.

2. Power Spectral Density Model

2.1 SCS Radio Emission

A string segment with tension $\mu = G\mu \cdot c^4/G$ and supercurrent I at frequency f produces:

$$P(f) = \frac{\mu \cdot I^2 \cdot f}{4\pi d^2} \cdot [\text{SCm}]_{\text{emit}}$$

Parameter	Fiducial Value	Description
$G\mu/c^2$	10^{-8}	String tension
I	10^{10} A	Macroscopic supercurrent
f	1.4 GHz	L-band radio frequency
d	3.086×10^{25} m	~1 Gpc

2.2 Flux Density

The observed flux density in Jansky (1 Jy = 10^{-26} W/m^2/Hz):

$$S = \frac{P(f)}{10^{-26}}$$

2.3 Burst Energy

For a millisecond-duration burst:

$$E_{\text{burst}} = P(f) \cdot \Delta t = P(f) \times 10^{-3}\,\text{J}$$

3. Frequency Spectrum

The model predicts increasing power with frequency ($P \propto f$), modulated by the $[\text{SCm}]$ emission factor:

Frequency	$P(f)$ (W/Hz)	S (Jy)	Detectable?
400 MHz	computed	computed	String-dependent
1.0 GHz	computed	computed	String-dependent
1.4 GHz	computed	computed	String-dependent
5.0 GHz	computed	computed	String-dependent
10.0 GHz	computed	computed	String-dependent

Detectability threshold: $S > 10$ mJy for current radio surveys (ASKAP, MeerKAT, CHIME).

4. FRB Production Mechanism

SCS cusps ($P \propto f^{-4/3}$ at high frequencies) and kinks ($P \propto f^{-5/3}$) produce broadband transient emission. The $[\text{SCm}]$ emission coefficient modulates the total radiated power:

$$P_{\text{total}} = \int_0^\infty P(f) \cdot [\text{SCm}]_{\text{emit}}\, df$$

The UQFF framework predicts that SCS-produced FRBs should exhibit: - Millisecond durations (cusp crossing time) - Broadband spectra from MHz to GHz - Linear polarisation from ordered magnetic fields along the string - No repetition (one-off events from string loop collapse)

5. Conclusions

Radio constraints on SCS emission are consistent with the UQFF $[\text{SCm}]$ emission model. The framework provides a natural FRB production mechanism from cosmic string cusps. CP4 class `SCSSpectralSignaturesRadioCalculator` (#618) implements frequency, supercurrent, distance, and $G\mu$ sweeps.

References

1. arXiv:2305.09816 (2023)
2. Murphy, D.T., Star-Magic UQFF Framework (2024–2026)